色散关系：

$$\frac{c^2k^2}{\omega^2} = 1 - \frac{\omega_p^2/\omega^2}{1-\omega_c/\omega} \qquad (R波)$$

$$\frac{c^2k^2}{\omega^2} = 1 - \frac{\omega_p^2/\omega^2}{1+\omega_c/\omega} \qquad (L波)$$

$$\omega^2 - c^2k^2 = \frac{\omega_p^2}{1 \mp (\omega_c/\omega)}$$

右旋

$$k = \sqrt{\left[\omega^2 - \frac{\omega_p^2}{1-(\omega_c/\omega)}\right]}/c$$

$$\frac{\partial k}{\partial \omega} = \frac{1}{c} \cdot \frac{1}{2\sqrt{\omega^2 - \frac{\omega_p^2}{1-\omega_c/\omega}}} \left(2\omega + \frac{\omega_p^2}{(1-\omega_c/\omega)^2} \frac{\omega_c}{\omega^2}\right)$$